

CSE488 – Big Data Analytics

Assignment: Exploring FAISS Indexing for Large-Scale Vector Search

1. Assignment Overview

In modern data-driven systems, we often need to search through extremely large collections of data efficiently. Many applications such as recommendation systems, semantic search, image retrieval, and AI assistants rely on finding similar items quickly in large datasets.

In this assignment, you will explore FAISS (Facebook AI Similarity Search), a powerful library designed for efficient similarity search on large vector datasets. You will experiment with several FAISS indexing methods and compare their performance in terms of speed, accuracy, and memory usage.

The main goal is to understand how different indexing strategies behave when working with large datasets.

2. Learning Objectives

After completing this assignment, you should be able to:

- Understand the concept of vector similarity search
- Use the FAISS library for indexing large datasets
- Implement multiple FAISS indexing methods
- Evaluate and compare performance of different indexes
- Analyze trade-offs between accuracy and search speed

3. Dataset Requirement

You will need to work with a dataset containing vector embeddings. **The dataset must contain at least 100,000 vectors.** Look into Kaggle and other similar websites for datasets.

Possible options include:

Option 1 – SIFT Dataset: A commonly used benchmark dataset containing 128-dimensional vectors.

Option 2 – Deep1M Dataset

A collection of multiple vector datasets for nearest neighbor search experiments can be found here: <https://ann-benchmarks.com/index.html>

Important:

The dataset should contain at least 100,000 vectors. Using larger datasets such as 500K or 1M vectors will help you observe more interesting performance differences.

No two members from the same group (for the term project) can choose the same dataset.

4. FAISS Indexing Methods to Implement

You must implement and test at least FOUR different FAISS indexing methods:

1. IndexFlatL2 (Exact search baseline) (kNN)
2. IndexIVFFlat (Inverted File Index)
3. IndexIVFPQ (IVF + PQ)
4. IndexHNSW (Graph-based)
5. IndexLSH (optional)

5. Experiment Setup

For each indexing method, you should perform the following steps.

Step 1 – Build the Index (Training)

Measure the following:

- Time required to build the index
- Memory usage of the index

Step 2 – Perform Search Queries (Inference)

Run approximately 100-1000 search queries and record:

- Average query time with standard deviation
- Total search time
- Top-k results (k = 10)

6. Performance Evaluation

Compare the indexes using the following metrics:

- Index build time
- Average query latency
- Recall@10 (accuracy compared with exact search)
- Memory usage
- Scalability when dataset size increases

7. Scalability Experiment

You must run experiments on at least TWO dataset sizes:

Example:

Dataset Size 1: 100,000 vectors

Dataset Size 2: 500,000 or 1,000,000 vectors

Discuss how performance changes as the dataset grows.

8. Visualization

You must include visual comparisons using graphs. Some suggested plots:

- Query time vs indexing method
- Recall vs search time
- Memory usage comparison

Use libraries such as Matplotlib or Seaborn to generate the plots.

9. Tools and Technologies

You may use the following tools:

- Python
- FAISS
- NumPy
- Pandas
- Matplotlib or Seaborn
- Google Colab or Jupyter Notebook

10. Submission Requirements

You must submit the following:

1. Source Code

Python notebook or scripts used to run the experiments.

2. Report (double-column, IEEE format, 4 pages maximum)

The report should include:

- Introduction
- Dataset description
- Explanation of indexing methods
- Experimental setup
- Results and analysis
- Graphs and visualizations
- Discussion and conclusions

I will ask questions about your report. So, make sure you know what you have generated.

11. Grading Rubric

Dataset preparation: 3 marks

Implementation of FAISS indexes: 5 marks

Experiment design and setup: 4 marks

Performance evaluation and analysis: 4 marks

Visualization and graphs: 2 marks

Report quality and clarity: 2 marks

Total: 20 marks

12. Bonus Opportunities (+5 marks)

You may receive bonus marks for:

- Running experiments on datasets larger than 1 million vectors
- Using GPU acceleration with FAISS
- Hyperparameter tuning (e.g., nlist, nprobe)
- Implementing additional indexing methods

Good luck and enjoy the holidays!